

7-1 Test Plan

This chapter aims to demonstrate that UPPMS works according to its functional and non-functional requirements. The test plan defines *what* is tested, *when*, with which tools, and who is responsible.

1) Test Scope

- Laravel API and Services layer (backend).
- University isolation rules (Multi-Tenancy / TenantScope).
- XML registration, proposals, academic tracks, committees, and genetic scheduling.
- Acceptance scenarios on Syrian Private University campus demo data (SPU Demo).

2) Test Environment

Item	Details
Test framework	PHPUnit 10 (Laravel)
Command	<code>php vendor/bin/phpunit</code> from backend <code>last/</code>
Database	MySQL (test connection via Laravel / <code>phpunit.xml</code>)
Frontend	Manual / acceptance testing on React SPA (Vite)

3) Schedule and Responsibilities

Phase	Owner	Activity
During each Sprint	Development team	Unit + Feature tests for the related feature
End of phase	Development team	Full suite run + Regression
Before delivery	Team + pilot users	Acceptance / User Testing on SPU Demo

4) Plan Outputs

- Documented test cases under `tests/Unit` and `tests/Feature`.
- Execution reports (pass / fail / duration).
- Quality metrics and user feedback.

Detailed test-case listings are completed in **Appendix C** for the final report submission.

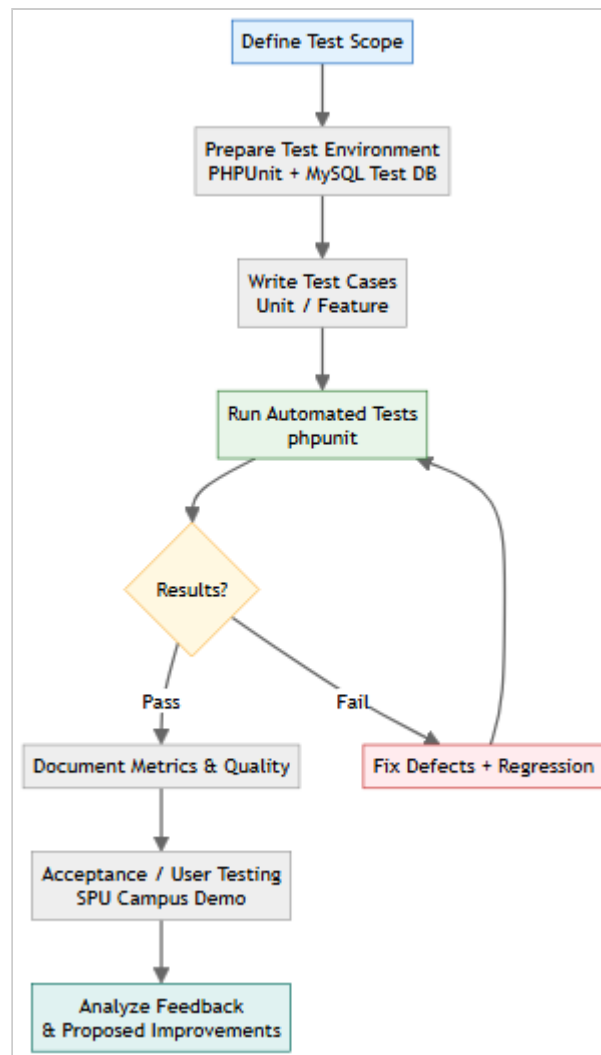


Figure 7-1: Test Plan Flow

Image file: diagrams-en/01-test-plan-flow.png